we imagine a billion years of low oxygen during Earth's middle chapters maintained by a mix of tectonic and biogeochemical controls. Foremost among those throttles were trace metal (Mo and V) deficiencies in the low-oxygen ocean and less efficient Fe nitrogenase — with the net effect of sustained limitations in fixed nitrogen that constrained primary production and corresponding oxygen release at low levels. What did it take to break the low-oxygen sustaining negative feedbacks? Our money is on a combination of first-order tectonic, evolutionary, and climatic controls woven together. Once tipped in the right direction, the result may have been a network of positive feedbacks that ultimately raised biospheric oxygen concentrations to a new, higher state. And now, thanks to the thought-provoking results of Sanchez-Baracaldo *et al.* [1], temporal patterns of evolution within the cyanobacterial lineage, particularly in the ocean, must be considered more carefully in that mix.

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Nutrition: Rejection Is the Fly's Protection

Animals need to ingest a full set of essential amino acids through their diet. A new study in *Drosophila* larvae describes how activation of the kinase GCN2 in three dopaminergic neurons mediates the rejection of amino-acid-imbalanced food.

Samantha L. Herbert and Carlos Ribeiro

The word homeostasis was coined in the early 20th century by Walter Cannon and popularized in his book *The Wisdom of the Body* [1]. The notion of nutrient homeostasis assumes that animals have an optimal nutrient intake, critical for both health and wellbeing, and that they should be able to select or reject food accordingly. The behavioral and physiological mechanisms underlying nutrient homeostasis are starting to be revealed and are an intense field of research within neuroscience. Whilst the senses, primarily smell and taste, are known to play an important role in food selection, it is becoming increasingly apparent that the 'quality control' of food continues after ingestion [2].

Studies on energy homeostasis found that animals can select metabolizable carbohydrates over non-metabolizable carbohydrates, independent of their sensory properties [3]. Furthermore, the formation of stable associative memories requires exposure to metabolizable carbohydrates, while the sensory properties of that sugar are secondary to the reinforcement [4–6].

Some of the earliest research on post-ingestive mechanisms focused on amino acid homeostasis. It was observed that amino-acid-deprived rodents rejected diets deficient in essential amino acids, and that post-ingestive amino-acid assessment involved a region of their olfactory cortex, known anatomically as the anterior piriform cortex (APC) [7]. A

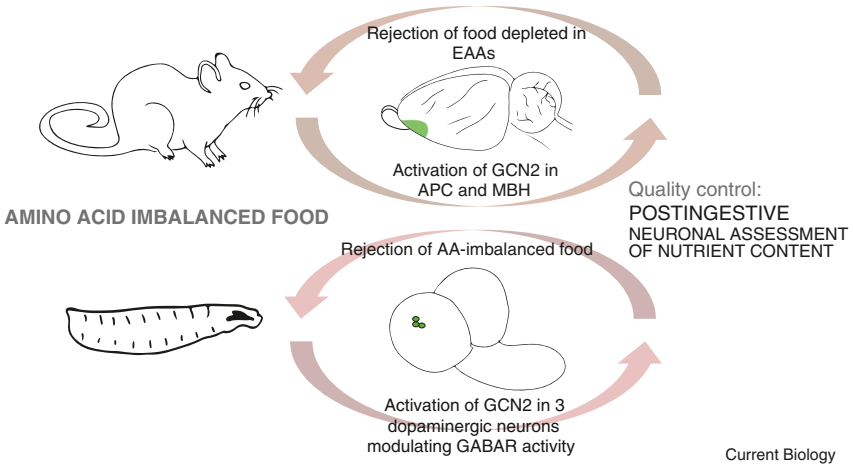


Figure 1. Post-ingestive quality control of dietary amino acids.

Top (brown arrows): in the rat, a diet deficient in essential amino acids increases the level of uncharged tRNAs, activating the GCN2 pathway in the anterior piriform cortex (APC, green) and the hypothalamus (MBH) [20]. This mechanism mediates the animal's response to reject the food. Bottom (pink arrows): as shown by Bjordal *et al*. [10], a similar mechanism operates in *Drosophila* larvae. Larvae reject an amino acid imbalanced diet in favor of increased wandering. In this case the GCN2 pathway acts in dopaminergic neurons (green) to regulate GABA signaling. AA, amino acids; EAA, essential amino acids.

conserved pathway mediating the rejection of amino-acid-imbalanced food was identified, involving the amino-acid-sensing protein kinase 'general control nonderepressing 2' (GCN2; Figure 1) [8,9]. These studies paved the way to address further key questions about nutrient assessment: which neuronal populations within the APC respond to amino-acid depletion; how does GCN2 transform nutrient sensing into a change in behavior; and how generalizable is the use of this nutrient-sensing mechanism in the post-ingestive neuronal assessment of food among the animal kingdom? In a recent study, Bjordal and colleagues [10] exploited the powers afforded by *Drosophila* neurogenetics to tackle these open questions.

In their work, Bjordal *et al.* [10] developed a new behavioral paradigm in *Drosophila* to test if larvae are sensitive to the amino acid content in food. They showed that, rather like rodents, fly larvae reject a diet of inferior amino-acid composition. Key to their experiments is a diet based on maize, a cereal low in amino acids, in particular two essential amino acids, tryptophan and lysine. When larvae are grown on a rich diet and then transferred to the maize-based diet, they start eating less, eventually leaving the food, in contrast to larvae transferred to a maize-based diet complemented with all amino acids. Given that these two behaviors are easy to score, they provide an ideal basis for dissecting the neuronal and molecular mechanisms underlying this nutritional response. The authors used an elegant combination of genetic amino-acid 'starvation' and genetic manipulations of neuronal activity to pinpoint the larval dopaminergic system as the circuit likely to mediate the rejection of amino-acid-imbalanced food. Furthermore, they show that, as in rodents, the activity of the GCN2 pathway in these cells is necessary and sufficient for rejection of the incomplete diet (Figure 1).

Importantly, Bjordal *et al.* [10] also performed live-imaging experiments to monitor the effect of diet on neuronal activity. Driving expression of the calcium indicator GCaMP3 in dopaminergic neurons in *ex vivo* brain preparations, they found that adding a full complement of amino acids had no effect on the calcium signal; however, the removal of multiple amino acids from the mix led to a highly reproducible and rapidly reversible increase in the calcium signal in specific subsets of dopaminergic neurons. The authors implicated the GCN2 pathway in the response by showing that knockdown of either GCN2 or activating transcription factor 4 (ATF4), a downstream component of the GCN2 pathway, in these neurons reduces their response to the absence of amino acids. This is a striking result, demonstrating that a specific set of neurons can respond not to the presence of nutrients, but rather to their absence.

This study is an impressive testament to the rapid progress now possible when combining behavioral readouts with the powerful neurogenetic tools available to *Drosophila* researchers. Bjordal *et al.* [10] managed to obtain evidence that GCN2-mediated rejection of imbalanced food is conserved in *Drosophila* larvae, and then went on to identify a small population of dopaminergic neurons as a likely substrate for reading out the nutrient content of the ingested diet, mediating the cessation of feeding and the subsequent initiation of foraging.

The concept of nutrient sensors is key when thinking about the post-ingestive assessment of food. Molecularly, the emerging picture from both vertebrate and invertebrate studies is that there are only a few, ancient nutrient-sensitive pathways which act in diverse cell types to read out nutrient levels with unique outcomes. This is highlighted by the fact that both GCN2 and 'target of rapamycin' (TOR), another key neuronal amino acid sensor, have been shown to coordinate cellular responses of yeast cells to nutrient shortages [11]. In yeast, these pathways reorganize metabolic processes; in multicellular organisms, however, they appear to have been coopted by the nervous system to also control the behavior and the physiology of the whole animal. This necessitated evolutionary changes allowing the pathways to have novel behaviourally relevant function. The finding of Bjordal *et al.* [10] that the neuronal GCN2 pathway regulates a GABA receptor to influence the activity of dopaminergic neurons illustrates this nicely [10].

One notable aspect of this study is that dopaminergic neurons react within seconds to the removal of amino acids, somewhat surprising

given the widely accepted view that GCN2 uses the presence of unloaded tRNAs as a proxy for the absence of amino acids [12]. If the dopaminergic neurons were to run out of charged tRNAs within a few seconds — the timescale of the neuronal response — it would result in a harmful block of protein translation. It is possible that GCN2 might be able to transform extremely small fluctuations in uncharged tRNAs into behavioral changes. Alternatively, the cell biology of nutrient-sensitive cells may be engineered to detect small fluctuations in metabolites, allowing them to react earlier than any other cell type to initiate a preemptive behavioral response. Glucose-sensing neurons, as well as insulin secreting beta cells, are good examples of such sensitized nutrient-responsive cells [13]. GCN2 has been shown to mediate stress signals other than amino-acid deprivation, so other upstream activating mechanisms are entirely conceivable too [12]. It will be important for future experiments to follow up this line of questioning, to help understand the molecular mechanisms that allow nutrient sensitive cells to perform their crucial task.

But what exactly is the larva rejecting and how? To answer this question we need a better understanding of the nutrient signal leading to the rejection of the diet. Is it the lack of an amino acid in the available food, or the individual proportions of these essential building blocks as suggested by the authors? Is it a change in the general nutritional state of the animal or a temporally localized signal induced by the short-term fluctuation induced by the ingested diet? The importance of these questions is highlighted by the fact that small changes in combinations of nutrients affect life history traits of the animal, such as growth, longevity and fertility [14,15], and that different dietary imbalances might be sensed by different nutrient sensing pathways. Recent papers describing chemically defined diets for flies, such as the Piper medium, might help in this line of investigation [15,16]. Finally, it will be important to clarify the exact behavioral response and how it is orchestrated by the observed changes in dopaminergic neurons. These neurons could, for example, trigger a change in the motor program of the larva or they could change the animal's perception of the food.

Compared to larvae which have to ingest as much food as possible to reach pupation, adult flies must carefully weigh up the opposing effects of amino acids on their fitness. In the adult fly, neuronal TOR signaling has been shown to modulate protein feeding [17,18]. Intriguingly, this pathway seems to play no role in larval post-ingestive amino acid assessment [10]. Whether GCN2 plays a similar role in adults, the interrelationship between both neuronal GCN2 and TOR nutrient sensitive pathways and how post-ingestive food assessment interacts with the readout of the internal metabolic state of the animal are important questions which need to be tackled next.

Insect physiology has made seminal contributions to our understanding of nutrient homeostasis [14,19]. Following in these footsteps, Bjordal *et al.* [10] have used the fly larva to explore the neuronal mechanisms at the basis of Cannon's *The Wisdom of the Body*. Importantly, their work shows that modern neuroscience has reached the maturity to tackle complex nutritional questions beyond pure energy homeostasis. How animals balance different macronutrients to achieve a healthy diet is one of the key questions in neuroscience and physiology. Given the impact of nutrition on health and wellbeing and the fact that protein homeostasis is likely to also be of great importance in humans [14], understanding the mechanisms mediating nutrient balancing is going to be a rewarding and highly relevant challenge.

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